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Roll No

AU/IP/IEM/PR/ME-403 (GS)
B.E. IV Semester Examination, June 2020
Grading System (GS)
Theory of Machines and Mechanisms
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. Explain the difference between mechanism and machine.
2. Define the following and give an example.
 - a) Kinematic link
 - b) Kinematic pair
 - c) Lower pair
 - d) Higher pair
3. The crank of a slider crank mechanism is 15cm and the Connecting rod is 60cm long. The crank makes 300rpm in the clockwise direction when it has turned 45° From the inner dead centre position determine:
 - i) Velocity of Slider
 - ii) Angular velocity of connecting rod and
 - iii) Linear velocity of the mid point of the connecting rod.
4. What is pressure angle in cam? Explain its significance.
5. What do you understand by the term Inversions? What are different inversions of four bar mechanism?
6. State and prove the law of gear tooth action for constant velocity ratio.

OR

What is a slider crank mechanism? Name all the four inversion of this mechanism and explain one of them in detail.
7. What do you mean by 'rubbing velocity at pin joints' ?

OR

Classify the different types of followers.
8. Write short notes on any two:
 - a) Stability of two wheeler
 - b) Kinematic pairs and their classification.
 - c) Ackermann's Steering mechanism
 - d) Synthesis of tooth profile for spur gears.
